

Lecture 4 : Limits

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1. Limit point of a set

Let S be a subset of $\mathbb{R}$. A point p in $\mathbb{R}$, is said to be a **limit point** (or an **accumulation point**, or a **cluster point**) of S , if every neighborhood of p contains a point of S other than p .

Therefore, p is a limit point of S , if for each positive number ϵ ,

$$[N(p, \epsilon) - \{p\}] \cap S \neq \emptyset.$$

$N(p, \epsilon) - \{p\}$, is called **deleted ϵ -neighborhood** of p .

Set of all the limit points of S , is said to be **derived set** of S and is denoted by S' .

Example Let $S = \mathbb{Q}$.

Since, $\mathbb{R} = \mathbb{Q} \cup \mathbb{Q}^c$ (where $\mathbb{Q}^c$ is set of all irrational number).

Let $p \in \mathbb{Q}$, then for any $\epsilon > 0$, $N(p, \epsilon)$ contains infinite number of rational as well as irrational points.

Therefore, $[N(p, \epsilon) - \{p\}] \cap \mathbb{Q} \neq \emptyset$. Thus, p is a limit point of S .

Since, p is an arbitrary rational point, therefore all rational points are limit point of S .

Let $q \in \mathbb{Q}^c$, then for any $\epsilon > 0$, $N(q, \epsilon)$ contains infinite number of irrational as well as rational points.

Therefore, $[N(q, \epsilon) - \{q\}] \cap \mathbb{Q} \neq \emptyset$. Thus, q is a limit point of S .

Since, q is an arbitrary irrational point, therefore all irrational points are also limit point of S .

Therefore, $S' = \mathbb{Q} \cup \mathbb{Q}^c = \mathbb{R}$.

2. Limit of a function

Let f be a real function defined on a domain $D \subseteq \mathbb{R}$. In order that, function f may have a limit l ($l \in \mathbb{R}$) at a point c , for x sufficiently close to c , $f(x)$ should be arbitrarily close to l . For this to be meaningful, it is necessary that c be a limit point of the domain D .

3. Definition of limit

Let $D \subseteq \mathbb{R}$ and $f : D \rightarrow \mathbb{R}$ be a function. Let c be a limit point of D . A real number l is said to be a limit of function f at c , if corresponding to any neighborhood V of l , there exists a neighborhood W of c , such that $f(x) \in V$ for all $x \in [W - \{c\}] \cap D$.

This can be expressed as $\lim_{x \rightarrow c} f(x) = l$.

4. $\delta - \epsilon$ definition of limit

Let $D \subseteq \mathbb{R}$ and $f : D \rightarrow \mathbb{R}$ be a function. Let c be a limit point of D . A real number l is said to be a limit of function f at c , if corresponding to a preassigned positive number ϵ , there exists a positive number δ , such that

$$l - \epsilon < f(x) < l + \epsilon \text{ for all } x \in N'(c, \delta) \cap D, \\ \text{where } N'(c, \delta) = \{x \in \mathbb{R} : 0 < |x - c| < \delta\} = (c - \delta, c + \delta) - \{c\}.$$

Theorem 1 *Let $D \subseteq \mathbb{R}$ and $f : D \rightarrow \mathbb{R}$ be a function. Let $c \in D'$ (D' is the set of all limit points of D). Then f can have at most one limit at c .*

Proof: Suppose, on the contrary, there exists two different limits, ' l ' and ' m ' of the function f at c .

Since, $l \neq m$ we assume $m > l$, without loss of generality. Let $\epsilon = \frac{m-l}{2} > 0$. Then the neighborhoods $(l - \epsilon, l + \epsilon)$ and $(m - \epsilon, m + \epsilon)$ are disjoint.

Since, l is a limit of function f at c , there exists a positive number δ_1 , such that

$$l - \epsilon < f(x) < l + \epsilon \text{ for all } x \in N'(c, \delta_1) \cap D.$$

Since, m is a limit of function f at c , thus there exists a positive number δ_2 , such that

$$m - \epsilon < f(x) < m + \epsilon \text{ for all } x \in N'(c, \delta_2) \cap D.$$

Let $\delta = \min\{\delta_1, \delta_2\}$.

Then,

$$l - \epsilon < f(x) < l + \epsilon \text{ and } m - \epsilon < f(x) < m + \epsilon \text{ for all } x \in N'(c, \delta) \cap D.$$

This is a contradiction, since the neighborhoods $(l - \epsilon, l + \epsilon)$ and $(m - \epsilon, m + \epsilon)$ are disjoint.

Therefore, $l = m$ and this completes the proof.

Example

1. Show that $\lim_{x \rightarrow 2} f(x) = 4$, where $f(x) = \frac{x^2-4}{x-2}$, $x \neq 2$.

Solution : Here, domain D of function f is $\mathbb{R} - \{2\}$ and 2 is a limit point of D .

$$\text{When } x \in D, |f(x) - 4| = \left| \frac{x^2-4}{x-2} - 4 \right| = |x - 2|.$$

Let us choose $\epsilon > 0$.

$|f(x) - 4| < \epsilon$ whenever $|x - 2| < \epsilon$ and $x \in D$, i.e. for all $x \in D$ satisfying $0 < |x - 2| < \epsilon$.

Therefore, $|f(x) - 4| < \epsilon$ for all $x \in N'(2, \delta) \cap D$ [taking $\delta = \epsilon$].

So, we have $\lim_{x \rightarrow 2} f(x) = 4$.

2. Show that $\lim_{x \rightarrow 0} f(x)$ does not exist, where $f(x) = [x]$.

Solution : Here, domain D of function $f(x) = [x]$ is $\mathbb{R}$.

Clearly, for every $\delta > 0$,

$$f(x) = \begin{cases} 0 & \text{when } x \in (0, \delta) \\ -1 & \text{when } x \in (-\delta, 0). \end{cases}$$

Let us choose $\epsilon = 1/2$. Then, for some $x, y \in N(0, \delta)$, we have $|f(x) - f(y)| = 1$. This implies $|f(x) - f(y)| > \epsilon$ for some $x, y \in N(0, \delta)$.

Thus, there does not exist any $l \in \mathbb{R}$ for function $f(x) = [x]$, such that $|f(x) - l| < \epsilon$ for all $x \in N(0, \delta)$.

Hence, $\lim_{x \rightarrow 0} f(x)$ does not exist at $x = 0$

Theorem 2 Let $D \subseteq \mathbb{R}$ and f and g are functions on D to $\mathbb{R}$. Let $c \in D'$ (D' is the set of all limit points of D) and $\lim_{x \rightarrow c} f(x) = l$, $\lim_{x \rightarrow c} g(x) = m$. Then

1. $\lim_{x \rightarrow c} (f+g)(x) = l+m$, where $f+g : D \rightarrow \mathbb{R}$ is defined by $(f+g)(x) = f(x)+g(x)$, $x \in D$.
2. $\lim_{x \rightarrow c} (f-g)(x) = l-m$, where $f-g : D \rightarrow \mathbb{R}$ is defined by $(f-g)(x) = f(x)-g(x)$, $x \in D$.
3. If $k \in \mathbb{R}$, $\lim_{x \rightarrow c} (kf)(x) = kl$, where $kf : D \rightarrow \mathbb{R}$ is defined by $(kf)(x) = k.f(x)$, $x \in D$.
4. $\lim_{x \rightarrow c} (f.g)(x) = lm$, where $f.g : D \rightarrow \mathbb{R}$ is defined by $(f.g)(x) = f(x).g(x)$, $x \in D$.

Proof:

1. We have,

$$\begin{aligned} |(f+g)(x) - (l+m)| &= |f(x) + g(x) - (l+m)| \\ &\leq |f(x) - l| + |g(x) - m|. \end{aligned}$$

Let us choose, $\epsilon > 0$.

Since, $\lim_{x \rightarrow c} f(x) = l$, so there exists a positive number δ_1 , such that

$$|f(x) - l| < \frac{\epsilon}{2} \text{ for all } x \in N'(c, \delta_1) \cap D.$$

Since, $\lim_{x \rightarrow c} g(x) = m$, so there exists a positive number δ_2 , such that

$$|g(x) - m| < \frac{\epsilon}{2} \text{ for all } x \in N'(c, \delta_2) \cap D.$$

Let $\delta = \min\{\delta_1, \delta_2\}$.

Then, $|f(x) - l| < \frac{\epsilon}{2}$ and $|g(x) - m| < \frac{\epsilon}{2}$ for all $x \in N'(c, \delta) \cap D$. Hence,

$$\begin{aligned} |(f+g)(x) - (l+m)| &\leq |f(x) - l| + |g(x) - m| \\ &< \frac{\epsilon}{2} + \frac{\epsilon}{2} \text{ for all } x \in N'(c, \delta) \cap D. \end{aligned}$$

This implies, $|(f+g)(x) - (l+m)| < \epsilon$ for all $x \in N'(c, \delta) \cap D$.

This proves that, $\lim_{x \rightarrow c} (f+g)(x) = l+m$.

2. We have,

$$\begin{aligned} |(f-g)(x) - (l-m)| &= |(f(x) - g(x)) - (l-m)| \\ &= |(f(x) - l) - (g(x) - m)| \\ &\leq |f(x) - l| + |g(x) - m|. \end{aligned}$$

Let us choose, $\epsilon > 0$.

Since, $\lim_{x \rightarrow c} f(x) = l$, so there exists a positive number δ_1 , such that

$$|f(x) - l| < \frac{\epsilon}{2} \text{ for all } x \in N'(c, \delta_1) \cap D.$$

Since, $\lim_{x \rightarrow c} g(x) = m$, so there exists a positive number δ_2 , such that

$$|g(x) - m| < \frac{\epsilon}{2} \text{ for all } x \in N'(c, \delta_2) \cap D.$$

Let $\delta = \min\{\delta_1, \delta_2\}$.

Then, $|f(x) - l| < \frac{\epsilon}{2}$ and $|g(x) - m| < \frac{\epsilon}{2}$ for all $x \in N'(c, \delta) \cap D$. Hence,

$$\begin{aligned} |(f-g)(x) - (l-m)| &\leq |f(x) - l| + |g(x) - m| \\ &< \frac{\epsilon}{2} + \frac{\epsilon}{2} \text{ for all } x \in N'(c, \delta) \cap D. \end{aligned}$$

This implies, $|(f-g)(x) - (l-m)| < \epsilon$ for all $x \in N'(c, \delta) \cap D$.

This proves that, $\lim_{x \rightarrow c} (f-g)(x) = l-m$.

3. We have,

$$\begin{aligned} |(kf)(x) - kl| &= |(kf(x)) - kl| \\ &= |k||f(x) - l|. \end{aligned}$$

Let us choose $\epsilon > 0$.

Since, $\lim_{x \rightarrow c} f(x) = l$, so there exists a positive number δ , such that

$$|f(x) - l| < \frac{\epsilon}{|k|} \text{ for all } x \in N'(c, \delta) \cap D.$$

Hence,

$$|(kf)(x) - kl| < |k| \frac{\epsilon}{|k|} \text{ for all } x \in N'(c, \delta) \cap D.$$

This implies, $|(kf)(x) - kl| < \epsilon$ for all $x \in N'(c, \delta) \cap D$.

This proves that, $\lim_{x \rightarrow c} (kf)(x) = kl$.

4. We have,

$$\begin{aligned} |(f.g)(x) - (lm)| &= |(f(x)g(x)) - (lm)| \\ &= |(f(x) - l)g(x) + l(g(x) - m)| \\ &\leq |f(x) - l||g(x)| + |l||g(x) - m|. \end{aligned}$$

Since, $\lim_{x \rightarrow c} g(x)$ exists, so there exists positive numbers B and δ_1 , such that

$$|g(x)| < B \text{ for all } x \in N(c, \delta_1) \cap D.$$

Let $k = \max\{B, |l|\}$. Then, $k > 0$ and

$$|(f.g)(x) - lm| < k \left\{ |f(x) - l| + |g(x) - m| \right\} \text{ for all } x \in N(c, \delta_1) \cap D.$$

Let us choose, $\epsilon > 0$.

Since, $\lim_{x \rightarrow c} f(x) = l$, so there exists a positive number δ_2 , such that

$$|f(x) - l| < \frac{\epsilon}{2k} \text{ for all } x \in N'(c, \delta_2) \cap D.$$

Since, $\lim_{x \rightarrow c} g(x) = m$, so there exists a positive number δ_3 , such that

$$|g(x) - m| < \frac{\epsilon}{2k} \text{ for all } x \in N'(c, \delta_3) \cap D.$$

Let $\delta = \min\{\delta_1, \delta_2, \delta_3\}$.

Then, $|(f.g)(x) - lm| < k \cdot \frac{\epsilon}{2k} + k \cdot \frac{\epsilon}{2k}$ for all $x \in N'(c, \delta) \cap D$.

This implies, $|(f.g)(x) - lm| < \epsilon$ for all $x \in N'(c, \delta) \cap D$.

This proves that, $\lim_{x \rightarrow c} (f.g)(x) = lm$.

Reference book : *Introduction to real analysis by S.K. Mapa.*